

10. Students used a quadrat to investigate the abundance of dandelions (*Taraxacum officinale*) on the school rugby pitch.



The method they used was:

- 1. Use a 1 m² quadrat.
- 2. Use a random number generator to place the quadrat on the rugby pitch.
- 3. Count the number of dandelions in the quadrat.
- 4. Repeat steps 2 and 3 another 5 times.
- 5. Calculate a mean.
- 6. Calculate the number of dandelions on the school rugby pitch.

The results they obtained are given in **Table 10.1**.

Table 10.1

Quadrat	Number of dandelions
1	3
2	5
3	2
4	7
5	15
6	6
Mean	



Examiner
only

- (a)

(i)

Draw a circle around the anomalous result in **Table 10.1**. [1]
- (ii)

Calculate the mean number of dandelions per quadrat and **write your answer in Table 10.1. Do not use the anomalous result in your calculation.** [2]

- (b)

The teacher said that he did not have confidence in the mean. Suggest why. [1]

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- (c)

The total area of the school rugby pitch is 7 350 m². Use your answer to part (a)(ii) to calculate the total number of dandelions on the school rugby pitch. [2]

Total number of dandelions on the school rugby pitch =

- (d)

State the name of the method the students should use to investigate the distribution of the dandelions across the pitch. [1]

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END OF PAPER

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